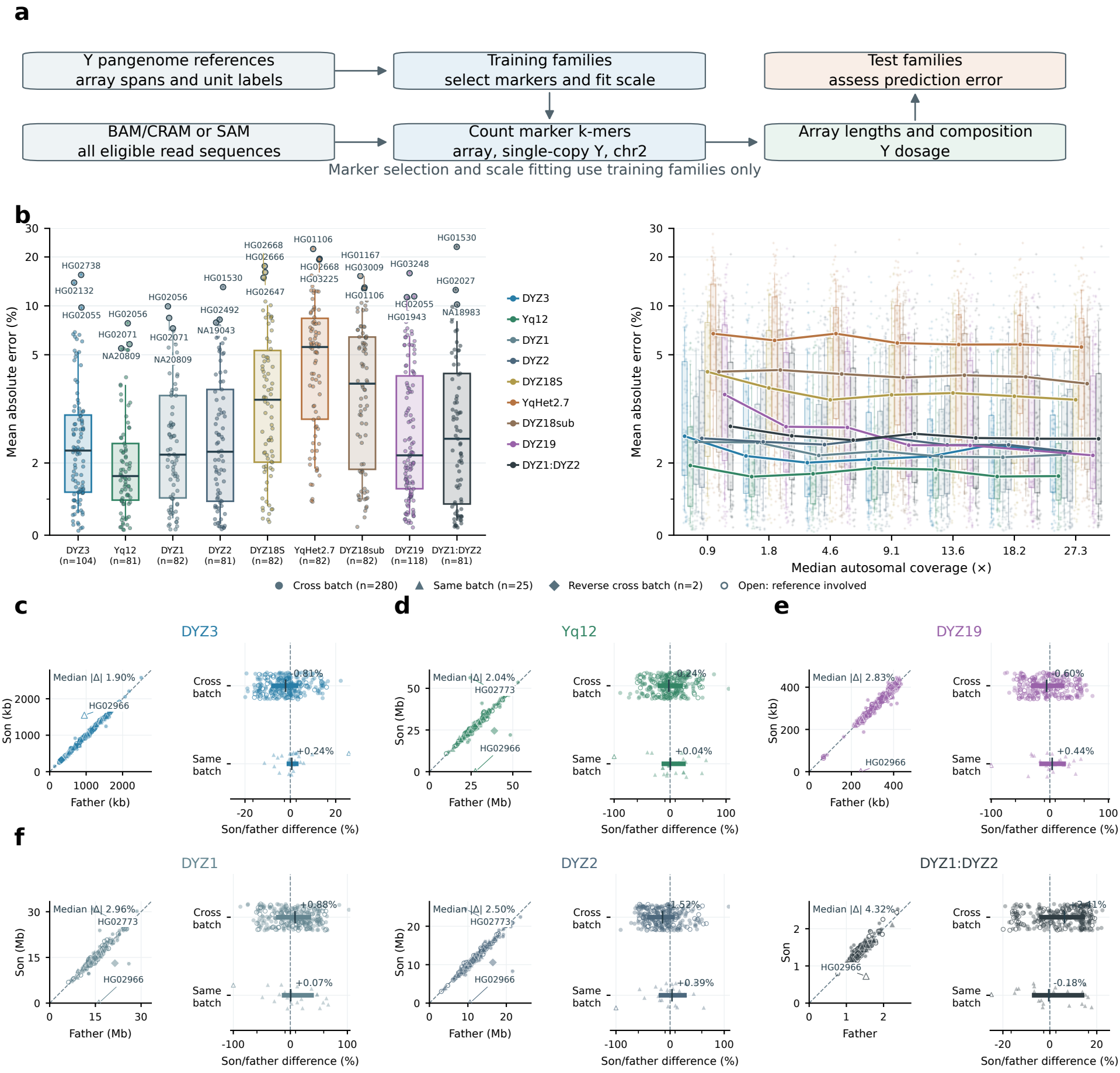

a
Y pangenome references
array spans and unit labels
Training families
select markers and fit scale
Test families
assess prediction error
BAM/CRAM or SAM
all eligible read sequences
Count marker k-mers
array, single-copy Y, chr2
Array lengths and composition
Y dosage
Marker selection and scale fitting use training families only
b
Mean absolute error (%)
DYZ3 (n=104)
Yq12 (n=81)
DYZ1 (n=82)
DYZ2 (n=81)
DYZ18S (n=82)
YqHet2.7 (n=82)
DYZ18sub (n=82)
DYZ19 (n=118)
DYZ1:DYZ2 (n=81)
DYZ3
Yq12
DYZ1
DYZ2
DYZ18S
YqHet2.7
DYZ18sub
DYZ19
DYZ1:DYZ2
Median autosomal coverage (×)
Cross batch (n=280)
Same batch (n=25)
Reverse cross batch (n=2)
Open: reference involved
c
DYZ3
Median |Δ| 1.90%
HG02966
Son (kb)
Father (kb)
Cross batch
Same batch
-0.81%
+0.24%
Son/father difference (%)
d
Yq12
Median |Δ| 2.04%
HG02773
HG02966
Son (Mb)
Father (Mb)
-0.24%
+0.04%
e
DYZ19
Median |Δ| 2.83%
-0.60%
+0.44%
f
DYZ1
Median |Δ| 2.96%
+0.88%
+0.07%
DYZ2
Median |Δ| 2.50%
-1.52%
+0.39%
DYZ1:DYZ2
Median |Δ| 4.32%
Son
Father
-2.41%
-0.18%